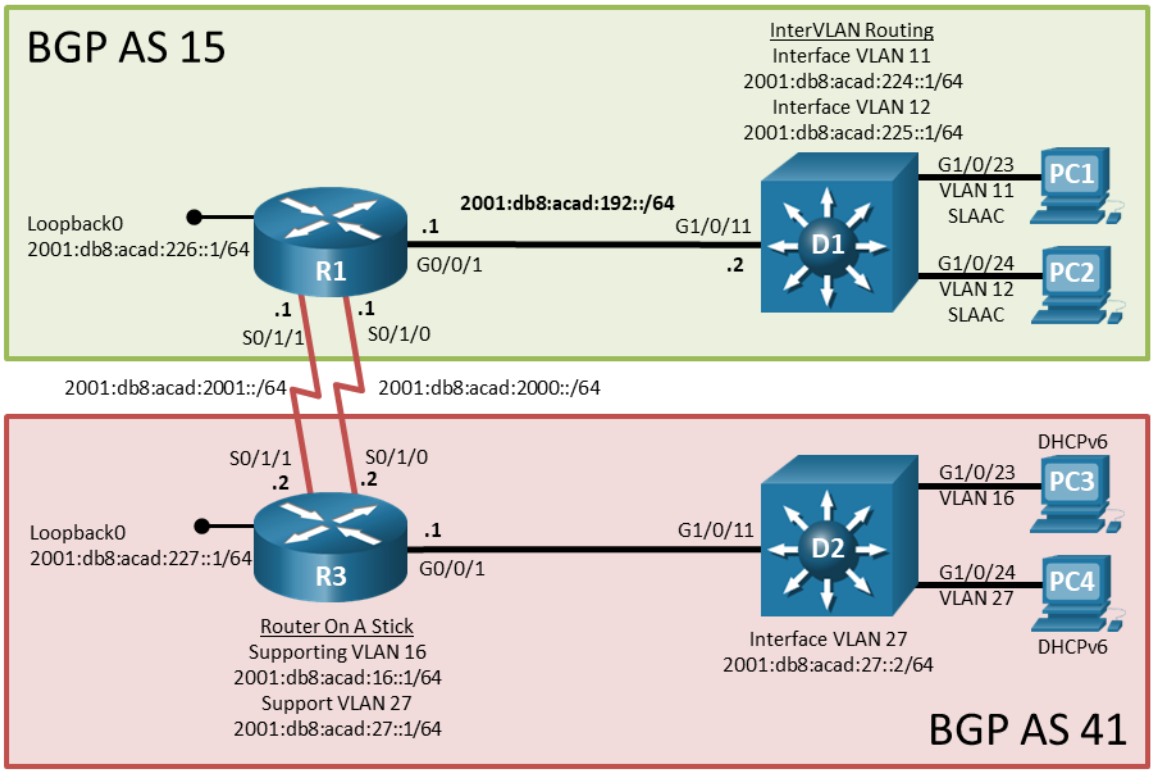
Lab - Troubleshoot IPv6 ACLs

Topology



Addressing Table

Device	Interface	IPv6 Address/Prefix Length	Link-Local Address
R1	G0/0/1	2001:db8:acad:192::1/64	fe80::1:1
	S0/1/0	2001:db8:acad:2000::1/64	fe80::1:2
	S0/1/1	2001:db8:acad:2001::/1/64	fe80::1:3
	Loopback 0	2001:db8:acad:226::1/64	fe80::1:4
R3	G0/0/1.16	2001:db8:acad:16::1/64	fe80::3:1
	G0/0/1.27	2001:db8:acad:27::1/64	fe80::3:2
	Loopback 1	2001:db8:acad:227::1/64	fe80::3:3
D1	G1/0/11	2001:db8:acad:192::2/64	fe80::d1:1
	VLAN 11	2001:db8:acad:224::1/64	fe80::d1:2

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Device	Interface	IPv6 Address/Prefix Length	Link-Local Address
	VLAN 12	2001:db8:acad:225::1/64	fe80::d1:3
D2	VLAN 27	2001:db8:acad:27::2/64	fe80::d2:1
	G1/0/11	2001:db8:1d1::2/64	fe80::d1:1
	Loopback 0	2001:db8:acad:1000::1/64	fe80::d1:2
	Loopback 1	2001:db8:acad:1001::1/64	fe80::d1:3
PC1	NIC	SLAAC	EUI-64/CGA
PC2	NIC	SLAAC	EUI-64/CGA
PC3	NIC	DHCPv6	EUI-64/CGA
PC4	NIC	DHCPv6	EUI-64/CGA

Objectives

Troubleshoot network issues related to the configuration and operation of IPv6 ACLs.

Background / Scenario

Your team operates a small enterprise lab that mirrors a regional campus network. R1 (ASN 15) and R3 (ASN 41) are peered via BGP using their loopback interfaces with multihop across the serial links between them. R1 maintains an OSPFv3 adjacency with distribution switch D1 and advertises a default route into the campus. R3 provides router-on-a-stick routing for user VLANs 16 and 27. Client segments behind D1 rely on SLAAC for IPv6 addressing, while client segments behind D2 are required to use DHCPv6 sourced from R3. Devices on VLANs 11 and 12 contain a large number of battery-powered Wi-Fi and IoT devices. To decrease energy consumption and channel contention, a policy was implemented to increase the periodic ICMPv6 Router Advertisement messages to 20 minutes. The clients will now rely on sending their ICMPv6 Router Solicitation message immediately after joining the network.

During a planned overnight change window, the night shift implemented IPv6 “hardening” controls, including several ACL updates at different enforcement points. Shortly after, multiple IPv6-related issues were reported by end users coming in for the morning shift.

You will be loading configurations with intentional errors onto the network. Your tasks are to FIND the error(s), document your findings and the command(s) or method(s) used to fix them, FIX the issue(s) presented here, and then test the network to ensure both of the following conditions are met:

1) the complaint received in the ticket is resolved
2) full reachability is restored

Note: The routers used with CCNP hands-on labs are Cisco 8200L with Cisco IOS XE Release 17.12.05c (universalk9 image). The switches used in the labs are Cisco Catalyst 9300L with Cisco IOS XE Release 17.12.05 (universalk9 image). Other routers, switches, and Cisco IOS versions can be used. Depending on the model and Cisco IOS version, the commands available and the output produced might vary from what is shown in the labs.

Note: Make sure that the devices have been erased and have no startup configurations. If you are unsure, contact your instructor.

Required Resources

- 2 Routers (Cisco 8200L with Cisco IOS XE Release 17.12.05c universal image or comparable)

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- 2 Switches (Cisco 9300L with Cisco IOS XE Release 17.12.05 universal image or comparable)
- 4 PCs (Choice of operating system with terminal emulation program installed)
- Console cables to configure the Cisco IOS devices via the console ports
- Ethernet and serial cables as shown in the topology

Part 1: Trouble Ticket 32.1.2.1

Scenario:

Following last night's "client VLAN lockdown" change, the help desk received reports that new or rebooted devices on VLAN 11 and VLAN 12 had long delays in obtaining IPv6 addresses via SLAAC, upwards of 20 minutes. Existing devices that did not renew may still appear online, but new joins and address refreshes fail. Your task is to assess the impact of the ACL changes on client autoconfiguration behavior and restore normal SLAAC operation without weakening the intended client access restrictions.

Be sure you are using devices that have been connected and prepared by your instructor. Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.2.1-r1-config.txt running-config`
R3	`copy flash:/enarsi/32.1.2.1-r3-config.txt running-config`
D1	`copy flash:/enarsi/32.1.2.1-d1-config.txt running-config`
D2	`copy flash:/enarsi/32.1.2.1-d2-config.txt running-config`

- PCs 1, 2, 3, and 4 should be configured for dynamic acquisition of an IPv6 address.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- After you have corrected the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #

- Save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the privileged EXEC command Trouble Ticket 32.1.2.2

Scenario:

Users on VLANs 16 and 27 behind R3 report they cannot reach internal gateway interfaces on D1 using IPv6 ping this morning, although routing adjacencies between R1 and R3 appear stable. The night shift notes indicate a "bogon/martian" IPv6 filter was introduced inbound on R1's serial links to prevent unrouted or inappropriate source addresses from entering ASN 15. It appears the new filter is too broad and is blocking legitimate campus prefixes. Your task is to adjust the control so that valid traffic from VLANs 16 and 27 can reach internal resources while preserving the intent of the filter.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.2.2-r1-config.txt running-config`
R3	`copy flash:/enarsi/32.1.2.2-r3-config.txt running-config`

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Device	Command
D1	`copy flash:/enarsi/32.1.2.2-d1-config.txt running-config`
D2	`copy flash:/enarsi/32.1.2.2-d2-config.txt running-config`

- PCs 1, 2, 3, and 4 should be configured for dynamic acquisition of an IPv6 address.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- Once you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the privileged EXEC command **reset.now**. This script will clear your configurations and reload the devices.

Part 2: Trouble Ticket 32.1.2.3

Scenario:

Shortly after another “UDP hardening” change on R3’s subinterfaces on VLAN 16, Users on VLAN 16 reported to the help desk that the point-of-sale (POS) systems on VLAN 16 are not appearing in the network’s DHCPv6 lease logs. These Windows-based POS terminals are part of a monitored group of devices whose network activity must be tracked for compliance and auditing purposes. While it does not matter which IPv6 address each POS system receives, it is critical that the address be assigned by the DHCPv6 server so that the lease information is recorded. The DHCPv6 server assigns addresses with an interface ID beginning with “feed:”. Currently, these devices are not obtaining any IPv6 global unicast addresses. Your task is to troubleshoot the issue and ensure that the POS systems receive their IPv6 addresses from the DHCPv6 server.

Use the commands listed below to load the configuration files for this trouble ticket:

Device	Command
R1	`copy flash:/enarsi/32.1.2.3-r1-config.txt running-config`
R3	`copy flash:/enarsi/32.1.2.3-r3-config.txt running-config`
D1	`copy flash:/enarsi/32.1.2.3-d1-config.txt running-config`
D2	`copy flash:/enarsi/32.1.2.3-d2-config.txt running-config`

- PCs 1, 2, 3, and 4 should be configured for dynamic acquisition of an IPv6 address.
- Passwords on all devices are **cisco12345**. If a username is required, use **admin**.
- Once you have fixed the ticket, change the MOTD on EACH DEVICE using the following command:

  banner motd # This is $(hostname) FIXED from ticket <ticket number> #
- Then save the configuration by issuing the **wri** command (on each device).
- Inform your instructor that you are ready for the next ticket.
- After the instructor approves your solution for this ticket, issue the privileged EXEC command **reset.now**. This script will clear your configurations and reload the devices.

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